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Sex differences in cognitive abilities: Analyses for the German WAIS-IV

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ABSTRACT

Sex differences on 15 subtests, the four indices, and the full scale IQ of the Wechsler Adult Intelligence Scale - Fourth Edition (WAIS-IV) are examined in the total national standardization sample ($N = 944$ males, 924 females) of the WAIS-IV. This sample, whose ages range from 16 to 89 years, were chosen by a stratified, random-sampling procedure to be representative of the population of Germany according to demographic features revealed in the 2012 census. Women scored lower in all indices and subtests with the exception of the Processing Speed Index (PSI) and the associated subtests, whereby the differences in means range between 3.2 and 4.9 IQ-points at the index level (FSIQ: Difference = 3.2, WMI: Difference = 4.9). But the maximum effect size of sex tends to be smaller ($\eta^2 = 0.050$) than the minimum effect size of education level (ranges from $\omega^2 = 0.055$ to $\omega^2 = 0.289$). It must be noted that the participants' education background, as defined by the highest attained education level, exerted a much greater effect upon cognitive abilities than sex did. A psychobiosocial model with the aim of accounting for sex-related differences in intelligence should be discussed.

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1. Introduction

The question as to whether there are sex differences in cognitive abilities has long been a matter of controversy. Several studies have come to the conclusion that there are no substantial sex differences in general intelligence (*g*-factor) (Camarata & Woodcock, 2006; Colom & Garcia-Lopez, 2002). While some authors propose an advantage in intelligence in favour of men conditioned by biological factors such as larger brain size in men (Lynn, 1994; Nyborg, 2005. Keith, Reynolds, Patel, and Ridley (2008) point out a slight yet significant advantage in general intelligence for girls and young women and found higher cognitive abilities in adulthood men only. Furthermore, it is well established that men have a higher variability in intelligence such that men are more strongly represented in the high and low ranges of the IQ-distribution than women (Johnson, Carothers, & Deary, 2008).

Traditionally, the idea of the hierarchical models of intelligence assumes that the differing tasks of a broad intelligence scale not only represents some share of the so-called *g*-factor but also an amount of task-specific intelligence (Spearman, 1904). Although this does not lead to a significant sex difference for the *g*-factor, sex differences play a role in regard to achievements on a subtest and index level (Halpern & LaMay, 2000). Research has shown that, despite some contradictoy results, men typically outperform women on of visual-spatial tasks, whereas women outperform men on test of processing speed (Keith et al., 2008).

By studying factor loadings, Colom, García, Juan-Espinosa, and Abad (2002) were able to demonstrate that there are no sex differences in general ability (*g*) on the Spanish standardization of the WAIS-III. They stated that the average sex differences favoring males have to be attributed to specific group factors and test specificity. Similarly, results obtained by Van der Sluis et al. (2006) based on the Dutch WAIS-III indicate differences between men and women in performance regarding specific cognitive abilities, but not in general intelligence (i.e. *g*). In contrast, for the US-standardization sample of the WAIS-III, Irwing (2012) reported sex differences not only in regards to specific abilities, but also in *g*. Men outperformed women in general intelligence (FSIQ) and on subtests like Information, Arithmetics and Symbol-Search, whereas women outperformed men on the Processing Speed Index (PSI).

The Wechsler tests in general examine a mix of cognitive abilities, nonetheless they show a male advantage on full scale IQ. Lynn (1994) traced significant differences in full scale IQ across different versions of the Wechsler scales. This effect is particularly apparent in adults.

All of the studies reviewed support the imperative for a detailed analysis of sex differences for each newly developed or adapted intelligence scale. Moreover, significant sex difference would have problematic consequences for testing as different norms would have to be applied to women and men. Thus there is, on both theoretical and practical grounds, need for clarification on the issue of sex differences in

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Table 1
Description of the WAIS-IV standardization sample (aggregated, with reference to German WAIS-IV Technical Manual).

Age group	Sex		Educational level			
	Male (%)	Female (%)	Low	Medium	High	n
16–17	50.0	50.0	18.8	36.1	45.1	122
18–19	50.0	50.0	4.5	18.1	77.3	88
20–24	50.0	50.0	13.3	33.5	53.2	158
25–29	50.0	50.0	13.1	33.1	53.8	130
30–34	50.0	50.0	14.7	37.3	48.0	102
35–44	50.0	50.0	22.7	45.4	31.8	154
45–54	50.0	50.0	28.1	41.0	30.8	156
55–64	47.4	52.6	43.0	29.9	27.2	114
65–69	48.4	51.6	33.9	33.8	32.3	124
70–74	44.1	55.9	43.2	34.7	22.0	118
75–79	39.7	60.3	65.4	12.8	21.8	78
80–84	33.3	66.7	54.2	27.1	18.8	48
85–90	33.3	66.7	42.4	27.3	30.3	33
Total	47.6	52.4				1425

Notes: Low = school not completed at all, special education, primary school (= up to 9 years of education); Medium = secondary school (= 10 years of education), High = grammar school or general qualification for tertiary study (12 or 13 years of education).

intelligence. In the present paper, this issue is addressed using the normative sample of the German edition of the Wechsler Adult Intelligence Scale - Fourth Edition (WAIS-IV, Petermann, 2012).

2. Material and methods

2.1. Sample

The analyses presented here are based on the standardization sample ($N = 1425$) of the German adaptation of the Wechsler Adult Intelligence Scale - Fourth Edition obtained in 2012. For reasons of representativeness, all participants were matched to the most recent census provided by the Federal Statistical Office on educational level and first language. The final sample consisted of n = 746 women (52.4%) and n = 679 men (47.6%), sampled all across the country. Further detail can be found in Table 1 as well as in the Technical Manual (Petermann, 2012). Education levels were assigned according to the following scheme: The category "low" includes those who did not complete school at all, or received special education, or attended primary school (up to 9 years of education); the "medium" category corresponds to secondary school (10 years of education), and the category "high" refers to the attendance at a grammar school or a general qualification for tertiary study (12 or 13 years of education).

2.2. Instruments

The WAIS-IV is an individually administered instrument for assessing intelligence among the age group 16–90. To obtain the four index scores and the Full Scale IQ (FSIQ), ten core subtests must be administered. Five supplemental subtests provide further information. The average administration duration is stated to range from 60 to 90 min. Appropriate reliability coefficients range from $r = 0.90$ (Processing Speed Index, PSI) to $r = 0.97$ (Verbal Comprehension Index, VCI) for index scores and $r = 0.98$ for the FSIQ. A detailed description of cognitive abilities measured by the subtests of the WAIS-IV is given in Table 2, that also matches WAIS subtests to the broad abilities of the Cattell-Horn-Carroll theory of cognitive abilities (CHC) (see Benson, Hulac, & Kranzler, 2010; Schneider & McGrew, 2012; Weiss, Keith, Zhu, & Chen, 2013) (Table 2).

2.3. Statistical methods

Statistical analysis was performed using SAS V. 9.4. Twenty two-way ANOVAs were used to compare the FSIQ, the four index scores, and the 15 subtest scores (factor 1: sex, factor 2: educational level). While the standard-ANOVA-table evaluates main and interaction effects by global F-test, simple effects as defined by Winer (1971) allow for a more detailed picture by analysing sex differences on each educational level. Aside from statistical significance, effect sizes (ω^2) and differences in

Table 2
Description of the WAIS-IV subtests in terms of the CHC-theory.

	Subtest	Abbr.	CHC	Cognitive abilities
VCI	Similiarities	SI	Crystallized knowledge (Gc), fluid reasoning (Gf)	Verbal reasoning and concept formation, auditory comprehension, memory, verbal expression, crystallized intelligence
	Vocabulary	VC	Crystallized knowledge (Gc)	Word knowledge, learning ability, long-term memory, degree of vocabulary development, crystallized intelligence
	Information	IN	Crystallized knowledge (Gc)	Crystallized intelligence, general factual knowledge, long-term memory
	Comprehension	CO	Crystallized knowledge (Gc)	
PRI	Block Design	BD	Visual processing (Gv)	Ability to analyze and synthesize abstract visual stimuli, nonverbal concept formation, visual perception and organization, visual-motor coordination, ability to separate figure-ground in visual stimuli
	Matrix reasoning	MR	Fluid reasoning (Gf)	Fluid intelligence
	Visual puzzles	VP	Visual processing (Gv)	Visual-spatial reasoning, understanding part-whole relationship, fluid intelligence, mental rotation, visual working memory, ability to analyze and synthesize abstract visual stimuli
	Figure weights	FW	Fluid reasoning (Gf)	Quantitative fluid reasoning and induction
	Picture completion	PC	Visual processing (Gv), crystallized knowledge (Gc)	
WMI	Digit span	DS	Short-term memory (Gsm)	Auditory short-term memory; cognitive flexibility, mental alertness, attention DS forward: auditory rehearsal, memory, focused attention DS backward and sequencing: working memory, mental manipulation, visualspatial imaging
	Arithmetic	AR	Fluid reasoning (Gf), Short-term memory (Gsm); quantitative knowledge (Gq)	Mental manipulation, concentration, focused attention, short- and long-term memory, numerical reasoning ability, applied computational ability
	Letter-number Sequencing	LNS	Short-term memory (Gsm)	Focused attention, concentration, auditory rehearsal, mental manipulation, working memory capacity
PSI	Symbol search	SYS	Processing speed (Gs)	Cognitive processing speed, visual short-term memory, visual-motor coordination, visual discrimination, concentration, decision-making speed
	Coding	CD	Processing speed (Gs)	Cognitive processing speed, visual short-term memory, incidential learning ability, visual perception, visual-motor coordination, visual scanning ability, attention, concentration
	Cancellation	CA	Processing speed (Gs)	Speed of visual-perceptual processing and decision making, visual scanning ability, attention, concentration

Abbrevations: VCI = Verbal Comprehension Index, PRI = Perceptual Reasoning Index. WMI = Working Memory Index, PSI = Processing Speed Index, Gc = Comprehension-Knowledge, Gf = Fluid Reasoning, Gv = Visual-Spatial Processing, Gs = Processing Speed, Gq = Quantitative Knowledge, Gsm = Short-Term Memory.

Table 3
Means and standard deviation of subtests, indices and FSIQ by educational level and sex.

EDL	Sex	N	FSIQ		GAI		VCI		PRI		WMI		PSI	
			M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
1		392	89.22	15.11	89.58	15.02	89.34	14.28	92.16	15.99	91.36	15.43	92.44	15.89
2		477	98.80	12.21	98.36	12.25	98.13	12.30	99.32	13.32	99.35	13.46	100.71	13.51
3		556	108.52	11.58	108.65	11.76	109.13	11.55	106.22	12.61	106.67	12.63	104.67	13.37
	m	679	101.61	15.35	102.09	15.22	101.75	14.97	102.12	15.18	102.62	15.72	98.26	14.71
	f	746	98.45	14.56	98.02	14.61	98.41	14.77	98.15	14.50	97.64	13.98	101.55	15.06
	Diff		3.16		4.07		3.34		3.97		4.98		−3.29	
1	m	186	89.94	16.00	91.16	16.07	90.38	14.17	93.92	17.52	92.65	16.66	89.87	15.14
	f	206	88.57	14.27	88.15	13.88	88.39	14.35	90.56	14.32	90.20	14.17	94.77	16.23
2	m	199	100.73	12.97	100.42	12.92	99.81	12.80	101.25	13.48	102.58	14.25	99.61	12.62
	f	278	97.42	11.45	96.88	11.54	96.93	11.80	97.94	13.05	97.04	12.37	101.51	14.08
3	m	294	109.60	10.90	110.12	10.83	110.26	11.12	107.89	11.85	108.95	12.48	102.66	13.54
	f	262	107.31	12.21	106.99	12.53	107.86	11.92	104.35	13.20	104.12	12.32	106.93	12.84

EDL	Sex	N	BD		SI		DS		MR		VC		AR	
			M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
1		392	8.77	3.19	8.24	2.90	8.47	3.04	8.49	3.09	8.03	3.05	8.53	3.07
2		477	9.96	2.82	9.76	2.71	9.99	2.68	9.95	2.74	9.67	2.67	9.81	2.74
3		556	11.14	2.78	11.55	2.60	11.07	2.67	11.19	2.60	11.88	2.41	11.29	2.60
	m	679	10.38	3.13	10.13	3.06	10.17	3.12	10.28	3.07	10.28	3.20	10.77	3.11
	f	746	9.83	2.97	9.96	3.01	9.83	2.82	9.81	2.90	9.90	3.02	9.37	2.73
	Diff		0.55		0.17		0.34		0.47		0.38		1.40	
1	m	186	9.01	3.37	8.12	2.85	8.45	3.13	8.72	3.41	8.03	3.10	8.99	3.35
	f	206	8.56	3.01	8.34	2.94	8.49	2.97	8.30	2.75	8.03	3.02	8.12	2.74
2	m	199	10.33	2.91	9.87	2.72	10.23	2.86	10.08	2.71	9.88	2.82	10.71	2.87
	f	278	9.69	2.72	9.68	2.71	9.81	2.54	9.86	2.77	9.52	2.56	9.17	2.45
3	m	294	11.28	2.80	11.56	2.62	11.22	2.79	11.40	2.59	11.97	2.45	11.93	2.52
	f	262	10.98	2.75	11.54	2.57	10.89	2.52	10.94	2.60	11.77	2.36	10.57	2.51

EDL	Sex	N	SS		VP		IN		CD		LN		FW	
			M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
1		392	8.93	3.17	8.93	2.98	8.11	2.98	8.29	3.11	8.34	2.89	8.22	3.34
2		477	10.13	2.81	9.83	2.80	9.74	2.51	10.11	2.67	9.84	2.60	9.54	2.64
3		556	10.62	2.75	10.82	2.81	11.58	2.42	11.04	2.70	11.16	3.01	11.11	2.57
	m	679	9.82	2.99	10.45	3.06	10.65	2.88	9.53	2.88	10.26	3.25	10.37	3.05
	f	746	10.14	2.94	9.53	2.77	9.42	2.92	10.38	3.08	9.91	2.82	9.50	2.89
	Diff		−0.32		0.92		1.23		−0.85		0.35		0.87	
1	m	186	8.62	3.13	9.35	3.38	8.80	2.87	7.68	2.91	8.23	2.92	8.54	3.52
	f	206	9.22	3.19	8.54	2.52	7.49	2.94	8.85	3.18	8.46	2.86	7.83	3.08
2	m	199	10.14	2.67	10.28	2.91	10.31	2.66	9.71	2.42	10.25	2.95	10.01	2.66
	f	278	10.12	2.90	9.51	2.67	9.33	2.31	10.39	2.80	9.54	2.26	9.19	2.58
3	m	294	10.37	2.89	11.26	2.70	12.06	2.23	10.57	2.56	11.34	3.09	11.58	2.41
	f	262	10.89	2.56	10.32	2.84	11.03	2.50	11.56	2.76	10.96	2.91	10.60	2.64

EDL	Sex	N	CO		CA		PC	
			M	SD	M	SD	M	SD
1		392	8.45	3.16	8.70	3.20	8.86	3.29
2		477	9.72	2.53	9.98	2.77	9.98	2.74
3		556	11.62	2.58	10.74	2.88	10.78	2.62
	m	679	10.37	3.15	9.92	3.14	10.16	2.80
	f	746	9.87	2.89	10.13	2.88	9.82	3.08
	Diff		0.5		−0.21		0.34	
1	m	186	8.38	3.19	8.71	3.32	9.16	3.08
	f	206	8.50	3.15	8.68	3.06	8.59	3.45
2	m	199	10.02	2.56	9.83	2.92	10.05	2.62
	f	278	9.51	2.49	10.09	2.65	9.94	2.82
3	m	294	11.87	2.70	10.63	2.99	10.87	2.51
	f	262	11.34	2.41	10.86	2.75	10.68	2.73

Abbrevations

AR	Arithmetics	BD	Block design
CA	Cancellation	CD	Coding
CO	Comprehension	DS	Digit Span
FW	Figure Weight	IN	Information
LN	Letter Number Sequencing	MR	Matrix Reasoning
PCm	Picture Completion	SI	Similiarities
SS	Symbol search	VC	Vocabulary
VP	Visual Puzzles		
FSIQ	Full Scale IQ	GAI	General Ability Index
VCI	Verbal Comprehension Index	PRI	Perceptual Reasoning Index

(continued on next page)

Table 3 (*continued*)

Abbrevations			
AR	Arithmetics	BD	Block design
WMI	Working Memory Index	PSI	Processing Speed Index
EDL 1	low educational level	sex m	male
EDL 2	medium educational level	sex f	female
EDL 3	high educational level		
M	Mean	SD	Standard Deviation

their respective metric (Δ) are reported (i.e. IQ-points on index level, scaled scores on subtest level).

3. Results

3.1. Sex differences in subtests and indices of the WAIS-IV

In Table 3 descriptive statistics (M, SD) are given for the subtests, composite scores as well as Full Scale IQ subset by sex and educational level. Substantial differences appear with regard to the three levels of education (low, medium, high). Looking, for example, at the FSIQ or the Verbal Comprehension Index (VCI), the mean difference between higher and lower education reaches up to 20 IQ-points (IQ metric: M = 100, SD = 15). For the subtests (scaled score metric: M = 10, SD = 3), differences range from 2 to 3 scaled scores with the greatest differences being found for verbal subtests. Women scored lower in all indices except for Processing Speed Index (PSI. Consistently higher standard deviations can be observed for men except for Processing Speed Index and the subtests Coding und Picture Completion.

Table 4 provides an inferential analysis of these means by ANOVA. Two of the nine significant differences are in favour of women. Note that, while there are significant main effects of both sex and educational level, not a single interaction term could be found on subtest, index or Full Scale IQ level.

Additional drill-down analysis was performed by slicing interactions into simple effects (that is: testing sex differences on each educational levels as well as EDL 2-df-differences for each sex) and testing between-EDL-differences within the sex effect. The highly significant effects of educational level were uniformly distributed among both sexes in any analyis (subtests, indices, FSIQ) and have therefore been dropped from Table 5, as have been all tests of whether sex differences per se were different between educational levels (all *failed* to reach significance), the full table may be downloaded from PAID Supplementary material online resources. However, the effects of sex did vary across educational levels. For the FSIQ, WMI and VCI, no effect of sex on performance could be observed within the lowest category of educational level (EDL 1). For the PSI, a weak effect was present only within the medium educational level. On the subtest level the results are

Table 4
Results of the analysis of variance for subtests, indices and the FSIQ of the WAIS-IV taking educational level and sex into account.

Scale	Effect	*df*	*F*	*p*	ω^2	$p \cdot \omega^2$	Scale	Effect	*df*	*F*	*p*	ω^2	$p \cdot \omega^2$
FSIQ	Sex	1	11.35	0.001	0.005	0.007	SS	Sex	1	5.54	0.019	0.003	0.003
	EDL	2	259.87	0.000	0.264	0.266		EDL	2	41.19	0.000	0.053	0.053
	Sex*EDL	2	0.62	0.539	0.000	0.000		Sex*EDL	2	1.59	0.204	0.001	0.001
GAI	Sex	1	21.94	0.000	0.011	0.014	VP	Sex	1	30.84	0.000	0.019	0.021
	EDL	2	253.87	0.000	0.258	0.262		EDL	2	49.15	0.000	0.062	0.063
	Sex*EDL	2	0.05	0.948	0.000	0.000		Sex*EDL	2	0.13	0.877	0.000	0.000
VCI	Sex	1	12.86	0.000	0.006	0.008	IN	Sex	1	64.48	0.000	0.033	0.043
	EDL	2	287.79	0.000	0.284	0.287		EDL	2	205.41	0.000	0.213	0.223
	Sex*EDL	2	0.14	0.873	0.000	0.000		Sex*EDL	2	0.52	0.594	0.000	0.000
PRI	Sex	1	21.15	0.000	0.012	0.014	CD	Sex	1	40.22	0.000	0.023	0.027
	EDL	2	117.18	0.000	0.138	0.140		EDL	2	117.62	0.000	0.138	0.141
	Sex*EDL	2	0.01	0.990	0.000	0.000		Sex*EDL	2	0.88	0.416	0.000	0.000
WMI	Sex	1	34.40	0.000	0.019	0.023	LN	Sex	1	2.69	0.101	0.001	0.001
	EDL	2	142.89	0.000	0.162	0.166		EDL	2	82.18	0.000	0.123	0.124
	Sex*EDL	2	1.50	0.223	0.001	0.001		Sex*EDL	2	2.11	0.122	0.002	0.002
PSI	Sex	1	23.89	0.000	0.014	0.016	FW	Sex	1	24.27	0.000	0.017	0.020
	EDL	2	91.19	0.000	0.111	0.112		EDL	2	96.47	0.000	0.140	0.143
	Sex*EDL	2	1.43	0.240	0.001	0.001		Sex*EDL	2	0.21	0.808	0.000	0.000
BD	Sex	1	8.53	0.004	0.005	0.005	CO	Sex	1	4.40	0.036	0.002	0.002
	EDL	2	75.26	0.000	0.094	0.094		EDL	2	159.98	0.000	0.181	0.182
	Sex*EDL	2	0.45	0.640	0.000	0.000		Sex*EDL	2	1.99	0.137	0.001	0.001
SI	Sex	1	0.00	0.996	0.000	0.000	CA	Sex	1	0.72	0.396	0.000	0.000
	EDL	2	173.87	0.000	0.195	0.195		EDL	2	40.45	0.000	0.065	0.065
	Sex*EDL	2	0.60	0.546	0.000	0.000		Sex*EDL	2	0.20	0.815	0.000	0.000
DS	Sex	1	2.48	0.115	0.001	0.001	PC	Sex	1	3.63	0.057	0.002	0.002
	EDL	2	99.61	0.000	0.121	0.122		EDL	2	50.75	0.000	0.065	0.065
	Sex*EDL	2	0.79	0.452	0.000	0.000		Sex*EDL	2	0.77	0.465	0.000	0.000
MR	Sex	1	5.94	0.015	0.003	0.003							
	EDL	2	105.17	0.000	0.127	0.128							
	Sex*EDL	2	0.25	0.775	0.000	0.000							
VC	Sex	1	1.66	0.198	0.000	0.000							
	EDL	2	240.79	0.000	0.251	0.252							
	Sex*EDL	2	0.49	0.614	0.000	0.000							
AR	sex	1	74.33	0.000	0.042	0.049							
	EDL	2	114.16	0.000	0.130	0.137							
	Sex*EDL	2	1.66	0.191	0.000	0.001							

Abbrevations: see Table 3.

Table 5
Simple effects.

Scale	Effect of sex at EDL	Δ	*F*	*df*	*p*	Scale	Effect of sex at EDL	Δ	*F*	*df*	*p*
FSIQ	1	1.36	1.11	1	0.293	AR	1	0.88	10.27	1	0.001
	2	3.31	7.76	1	0.005		2	1.54	37.47	1	0.000
	3	2.29	4.43	1	0.035		3	1.35	34.56	1	0.000
GAI	1	3.01	5.40	1	0.020	SS	1	-0.60	4.23	1	0.040
	2	3.54	8.89	1	0.003		2	0.03	0.01	1	0.924
	3	3.13	8.28	1	0.004		3	−0.52	4.54	1	0.033
VCI	1	1.99	2.45	1	0.118	VP	1	0.82	8.17	1	0.004
	2	2.88	6.11	1	0.014		2	0.77	8.64	1	0.003
	3	2.40	5.06	1	0.025		3	0.95	15.52	1	0.000
PRI	1	3.36	5.82	1	0.016	IN	1	1.31	25.71	1	0.000
	2	3.31	6.72	1	0.010		2	0.98	17.03	1	0.000
	3	3.55	9.21	1	0.002		3	1.02	22.22	1	0.000
WMI	1	2.45	3.19	1	0.074	CD	1	−1.17	17.50	1	0.000
	2	5.55	19.45	1	0.000		2	−0.68	6.96	1	0.008
	3	4.83	17.65	1	0.000		3	−0.99	17.55	1	0.000
PSI	1	-4.91	11.94	1	0.001	LN	1	−0.23	0.41	1	0.520
	2	−1.90	2.12	1	0.145		2	0.71	6.23	1	0.013
	3	−4.27	12.83	1	0.000		3	0.38	2.25	1	0.134
BD	1	0.44	2.27	1	0.132	FW	1	0.71	4.16	1	0.042
	2	0.64	5.57	1	0.018		2	0.82	8.80	1	0.003
	3	0.29	1.39	1	0.239		3	0.98	15.66	1	0.000
SI	1	−0.22	0.62	1	0.433	CO	1	−0.12	0.20	1	0.656
	2	0.19	0.59	1	0.442		2	0.51	4.10	1	0.043
	3	0.02	0.01	1	0.934		3	0.53	5.31	1	0.021
DS	1	−0.04	0.02	1	0.889	CA	1	0.03	0.005	1	0.945
	2	0.42	2.62	1	0.105		2	−0.26	0.778	1	0.378
	3	0.33	1.93	1	0.165		3	−0.23	0.743	1	0.389
MR	1	0.42	2.21	1	0.137	PC	1	0.57	3.90	1	0.048
	2	0.22	0.72	1	0.397		2	0.11	0.19	1	0.664
	3	0.46	3.75	1	0.053		3	0.19	0.63	1	0.429
VC	1	0.00	0.00	1	0.995						
	2	0.36	2.11	1	0.146						
	3	0.20	0.75	1	0.385						

Abbrevations: see Table 3.

more diffuse. Table 6 displays the statistically significant results of simple effects of sex by EDLin a simplified fashion.

4. Discussion

4.1. Comparison with other studies

First, it must be noted that educational background, as defined by the highest attained education level, explains far more variance in cognitive abilities than sex. This finding is consistent with the analyses on the German standardization of the Wechsler Intelligence Scale for Children-IV regarding school-form differences in cognitive abilities related to the German education system (Daseking, Lipsius, Petermann, & Waldmann, 2008).

Based on the present data, higher mean values for men can be observed in several areas, along with a greater variability of values among men. Johnson et al. (2008) reported similar results. The same seems to hold true for associated scholarly achievements in natural science subjects (Lynn & Mikk, 2008), with differences between boys and girls dissolving over the last several years in line with gender harmonization of interest in natural sciences.

Table 6
Sex differences by education level (simple effects as 1-df-contrasts).

Subtests/*Indices*	Education level		
	EDL 1	EDL 2	EDL 3
AR, CD, IN, FW, VP, ***GAI, PRI, PSI***	+	+	+
SS, ***PSI***	+	n.s.	+
BD, CO, ***FSIQ, WMI, VCI***	n.s.	+	+
PC	+	n.s.	n.s.
LN	n.s.	+	n.s.
DS, MR, SI, VC, CA	n.s.	n.s.	n.s.

Abbrevations: see Table 3.

Examining significant sex-related differences, men achieved higher scores in the domains *visual processing, fluid reasoning* and *verbal comprehension*, while women outperformed men in the *processing speed* related tasks. Inconsistent results were however recorded within the verbal domain: while significant differences in favor of male participants were observed in the Information and Comprehension subtests, differences in the Similarities and Vocabulary subtests proved insignificant. These two subtest focus primarily on lexical knowledge. Men's advantages in scores in comprehension knowledge were also reported by Camarata and Woodcock (2006) in their comprehensive study regarding sex differences using the Woodcock-Johnson tests.

No significant sex-related differences were observed within the subtests pertaining specifically to the capabilities of working memory and short-term memory (WMI: Digit Span, Letter Number Sequencing). Men also performed significantly better in the subtest Arithmetics (WMI), that, in addition to short-term memory capability, relates to calculating skills as well as fluid reasoning. Similar performance differences in calculating tasks were produced in other studies (Geary, Saults, Liu, & Hoard, 2000). In a study regarding sex-related differences in the WAIS-IV and the WMS-IV, Lepach and colleagues (Lepach, Reimers, Pauls, Petermann, & Daseking, 2015) were able to detect a significant advantage for women in the WMS-IV subtests for auditory memory performance, while no differences could be found in visuospatial memory performance. Both Digit Span and Letter Number Sequencing tasks place demands not only upon verbal (auditory) aspects (memorising and reproducing digits and letters), but also on visuospatial processing (arranging in correct order, mental rotation). This may help explain why no sex-related differences were recorded for these two subtests.

On average, female participants displayed better performance in the domain of processing speed on the German version of the WAIS-IV. These findings replicate other studies' results (Camarata & Woodcock, 2006; Roivainen, 2011). The contents of the task subsumed under the

Processing Speed Index relate to the capability to solve low level cognitive tasks under time pressure while retaining attention and concentration (Flanagan, McGrew, & Ortiz, 2000). It seems that males are less capable to effectively solve such tasks (working quickly under time pressure) than females are. An important facet within these tasks lies with fast scanning of character strings, which might provide some explanation of female performance advantage in the area of reading (Lynn & Mikk, 2009).

Performance advantages among girls in processing speed could also be replicated for the German version of the WISC-IV (Goldbeck, Daseking, Hellwig-Brida, Waldmann, & Petermann, 2010). The study in question also observed advantages among boys within the Verbal Comprehension Index and Perceptual Reasoning Index.

4.2. Possible causes for sex differences

There has been a considerable debate over how to explain sex-related differences in cognitive abilities. Besides biological factors, environmental factors such as parental care, education, and other psychosocial variables frequently re-enter the discussion (for a summary see Halpern, 2013; Zimmermann & Spangler, 2001). Genes located on the X-chromosome are thought to be linked to this variability too (Johnson, Carothers, & Deary, 2009). Halpern and colleagues propose a psychobiosocial model (Halpern & LaMay, 2000). Moreover, national settings can also have an impact on sex-related differences, as can be seen from comparative international studies for children and adolescents such as the TIMMS or PISA (see i. a. Lynn & Mikk, 2008).

5. Conclusions

If Full-Scale IQ is considered the best proxy for factor *g*, a statistically significant difference of 3.2 IQ points can be observed in favour of the male participants in the German WAIS-IV. These results are consistent with the findings of other studies, but, obviously, one also has to go into the practical side.

The present results might revive the long standing discussion whether or not testing cognitive abilities must rely on gender specific norm tables. Further research has to investigate into the nature of sex differences as regards both test theory and test production or application: Are we facing *bias* (introduced by item or sample selection) or a genuine *effect* (explainable by biological or sociological conditions or the interaction of both), and how can we transfer respective insight into more valid tests?

6. Limitations

The present study relies on measured variables (subtest) and derived scores (indices/FSIQ) in terms of means and standard deviations and does not account for sex-related differences for the latent domain (*g*, factor scores).

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.paid.2017.04.003.

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